
A Review of Psychometrics with AI Foundation Models

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Abstract

Psychometrics is a subfield of psychology concerned with the objective measurement of latent constructs, such as values, personality, mental disorders, and intelligence. AI Foundation models (FMs), notably large language and multimodal models, are reshaping this field by capturing rich linguistic and behavioural regularities at scale. This paper reviews the emerging synergy between FMs and psychometrics. First, we map practical applications of FMs across the measurement pipeline. Next, we describe key methodologies for enhancing FM performance in psychometric contexts via prompt engineering, fine-tuning, and agentic frameworks. We then examine the theoretical implications of FMs for psychometrics. Finally, we chart risks and offer actionable recommendations for the effective, rigorous, and ethical implementation of FMs in psychometric research and practice. Realizing the potential of FMs in psychometrics demands interdisciplinary collaboration and continued alignment between advances in computational and measurement sciences.

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1 Introduction

Psychometrics is a scientific discipline concerned with quantifying latent psychological constructs such as values, personality, and cognitive abilities [85, 115]. It widely supports crucial applications across education, clinical assessment, organizational decision-making, social policy, and beyond.

Psychometrics usually involves mapping responses from instruments (e.g., self-report questionnaires, structured interviews, and performance tasks) onto statistical models, and yields interpretable estimates of psychological constructs [15]. Instrument development has traditionally relied on iterative cycles of item drafting, expert review, and empirical testing to establish reliability and validity [96]. This principled workflow is costly and slow, and it struggles to keep pace with research and applications that are expanding in scope, complexity, and diversity [33]. Computational approaches have offered new leverage that promise gains in efficiency, precision, and scalability [146]. For instance, supervised natural language processing (NLP) models enable automated scoring of digital traces [64]; topic models and distributional semantics facilitate content analysis [144]; and knowledge tracing and cognitive diagnosis support adaptive testing [8].

Earlier computational and machine learning (ML) models used in psychometrics are mostly discriminative and task-specific. They are trained on annotated datasets to classify or predict outcomes within narrow domains. A profound shift has occurred with the advent of *AI foundation models (FMs)*, such as *large language models (LLMs)* [1, 7, 14, 44, 138]. They differ remarkably from prior computational approaches in their deep semantic understanding, strong generative capabilities, and generality. Whereas earlier models typically rely on shallow features such as bag-of-words, FMs encode rich, contextualized semantic representations. They allow the models to better capture latent psychological constructs and handle ambiguity with human-like interpretation. Second, beyond purely discriminative tasks, many FMs can generate multimodal and human-like content. This enables new possibilities such as respondent simulation. Third, FMs are versatile across diverse tasks via instruction following and in-context learning. They can readily perform new tasks simply by following user instructions, without additional training.

Therefore, FMs promise to systematically transform the psychometric pipeline. Recent advances enable more accurate content analysis [106], introduce novel forms of automation [158], provide generalist assessment tools [160], support innovative measurement formats [155], and position AI itself as a subject of measurement [157]. At the same time, this shift introduces new challenges. For instance, additional sources of variance (prompting, decoding strategies, and model version) now interact with human variability [9, 118]. As a result, new controls and new forms of validity evidence (stability under paraphrase, cross-model agreement, and alignment with human data) are needed [157, 172]. The expanded capabilities and broader applications of FMs also warrant careful consideration of their limitations, as well as their ethical and social implications [18, 147].

This review charts the emerging integration of FMs into psychometrics (Fig. 1). We begin by surveying the applications of FMs to psychometrics (§ 2). Then, we examine methods for adapting and enhancing these models for psychometric tasks (§ 3). Next, we explore the theoretical implications that FMs pose for psychometrics (§ 4). To support responsible research and deployment, we offer recommendations for rigorous evaluation and ethical use (§ 5). Throughout, an interdisciplinary perspective is adopted to bridge advances in computational methods and measurement science, ensuring broad accessibility for a cross-disciplinary audience. Key terminologies are summarized in Appendix A.

2 Applications of Foundation Models in psychometrics

FMs are increasingly being integrated into psychometric pipelines. This section reviews these applications and organizes them according to the key stages of the measurement pipeline. We illustrate these applications in Fig. 2 and provide a detailed summary with representative examples in Table 1.

2.1 Construct modeling and instrument development

Psychometric instrument development often begins with rigorous construct conceptualization. Then, it proceeds through operationalization, item generation, and validation. FMs are now being used to

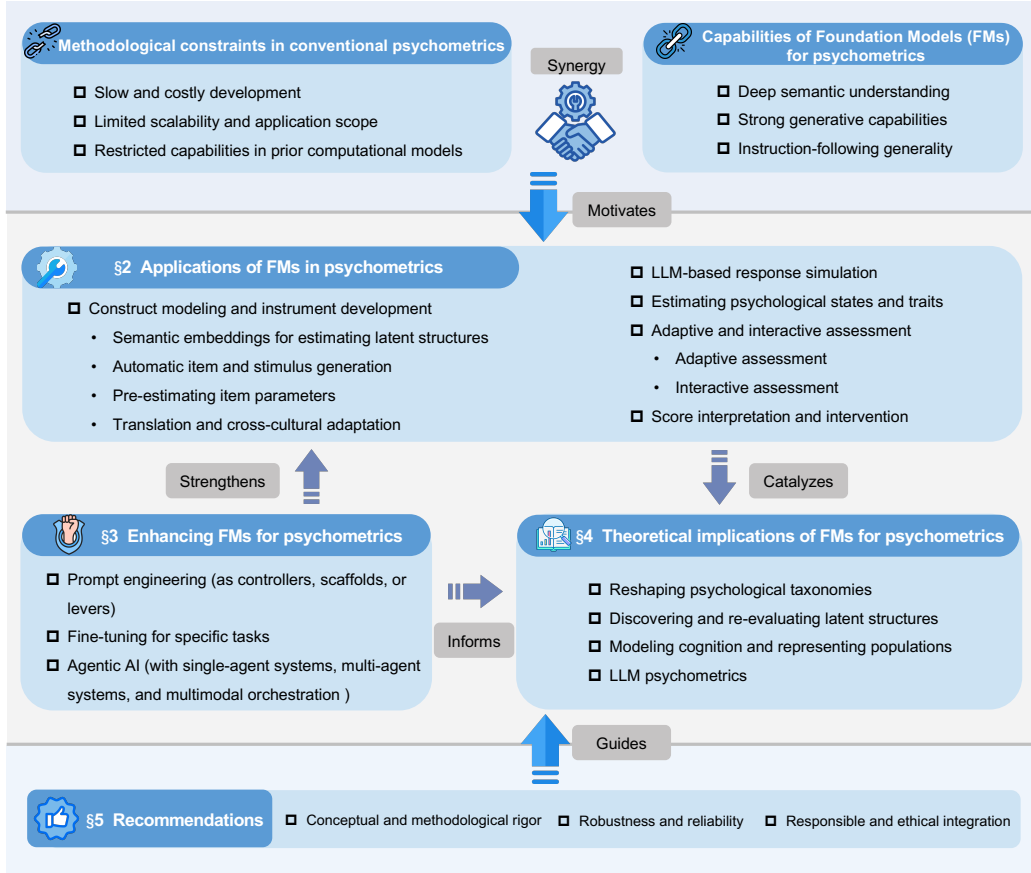


Figure 1: The structure of this review.

support this process holistically. Semantic embeddings help map and diagnose the construct space and reveal taxonomic issues such as jingle-jangle fallacies (§ 2.1.1). LLMs assist blueprint-constrained item and stimulus generation (§ 2.1.2). Models can pre-estimate item parameters to triage large item pools prior to piloting (§ 2.1.3). Also, AI systems aid translation and cross-cultural adaptation (§ 2.1.4).

2.1.1 Semantic embeddings for estimating latent structures

Semantic embeddings are numerical vectors that represent multimodal content in a high-dimensional space, where geometric proximity approximates semantic relatedness. These modern representations capture nuanced, context-dependent meaning, and can be compared using simple metrics such as cosine similarity.

Recent work shows that embedding similarity between test items can meaningfully predict their psychometric structure. For example, it predicts a priori factorial structures of personality tests with moderate-to-high correlation to human-derived structures [97]. For some scales, embedding similarity is shown to correlate with empirically observed item-item correlations ($r = 0.67$), and can help flag potential sources of model misfit [32] or abbreviate scales [63]. Principal component analyses (PCA) of text embeddings are shown to explain more cumulative variance than PCA of human responses [97]; some studies report near-perfect consistency between factor loading matrices derived from embeddings and from survey ratings [26]. These findings echo pre-LLM evidence that semantic relationships can account for stable portions of variance and, in some instruments, large fractions of observed item-item correlations [5].

LLM embeddings also enable extrapolation and integration. Research scales up analyses of personality lexicals with embeddings and provides evidence supporting two- or three-factor models of personality

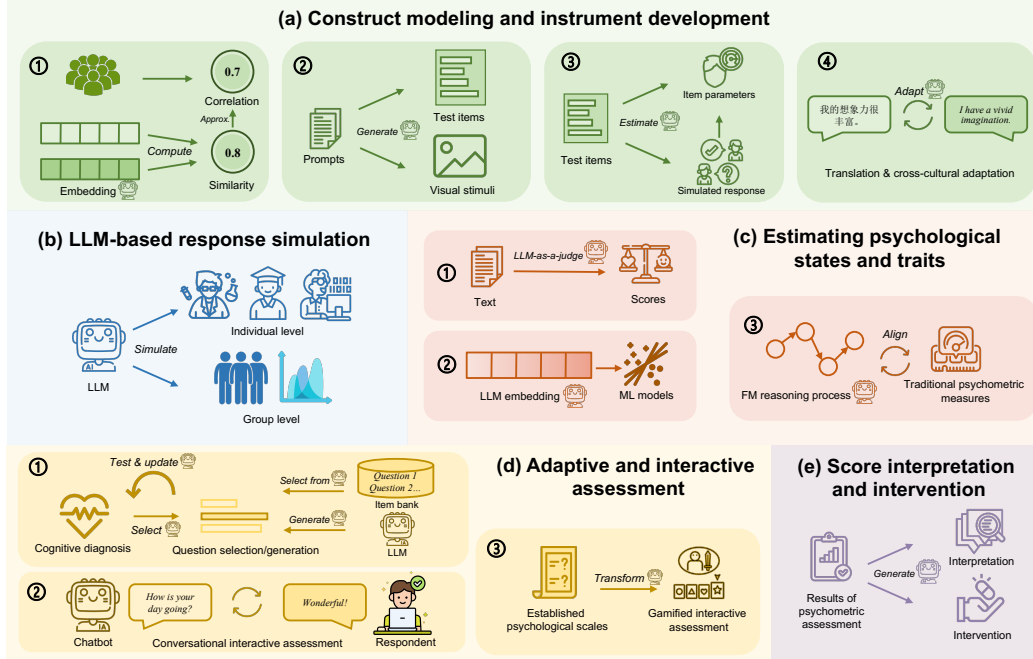


Figure 2: Overview of foundation model applications in psychometrics.

[26], in contrast to the more prominent big five model [38]. Interestingly, applying singular value decomposition (SVD) to LLM log-probabilities of trait-descriptive adjectives has also been reported to rediscover the big five traits [132]. These discrepancies are likely due to the sensitivity of LLMs to implementation choices [9]. At a broader level, mapping items, scales, and construct labels into a shared semantic space offers a scalable method for diagnosing potential taxonomic problems, such as the "jingle-jangle" fallacies that have long challenged the field [52, 151]. The theoretical implications of this capability are discussed further in § 4.

Semantic embeddings extend beyond text to encompass multimodal content. AI researchers utilize multimodal embeddings [113] to quantify the similarity between images and textual construct labels, which allows them to evaluate the value orientations of multimodal FMs [76].

Overall, embeddings provide powerful pre-data diagnostics and serve as a valuable scaffold for construct modeling and scale design. However, they should be viewed as a complement—not a substitute—for empirical validation and theoretically grounded interpretation. For example, research highlights the potential of human-AI collaboration in assessing item content validity [98]. It is important to note that semantic similarity may capture surface features such as wording or stylistic choices rather than underlying latent constructs, which can artificially inflate inter-factor correlations [31, 72]. Such deficiencies become apparent when embeddings fail to yield consistent results for reverse-scored items [63].

2.1.2 Automatic item and stimulus generation

The development of psychometric items and stimuli—ranging from test questions to evocative images—has long relied on labor-intensive processes and domain expertise. Automatic item generation (AIG) promises to alleviate this burden [36]. FMs have rapidly shifted AIG from hand-crafted templates to scalable, psychometrics-aware, and blueprint-constrained drafting with minimal human intervention.

Initial applications in language assessment demonstrate FMs capable of generating complete reading-comprehension test batteries [6]. This approach has since expanded to cognitive reasoning tasks, including natural language inference. Strategically prompted LLMs produce items with psychometric properties comparable to expert-authored baselines [71]. The scope has broadened beyond multiple-choice formats to encompass situational judgment tests and open-ended stimuli for personality and

Table 1: Overview of how FMs are used in psychometrics.

Application Area	Description & Key Benefits	Representative Examples
Semantic embeddings for internal structure & diagnostics (§ 2.1.1)	Numerical representations capture semantic relationships between items, predict factorial structures, and enable cross-measure comparisons.	Embedding similarity predicts item-item correlations and factorial structures [97]; Mapping constructs into shared semantic space reveals jingle-jangle fallacies and addresses taxonomic incommensurability [151].
Automatic item & stimulus generation (§ 2.1.2)	Scalable creation of psychometrically sound test items, from multiple-choice questions to visual stimuli.	Creative and personality assessment items with retained psychometric properties [60, 73]; Emotionally evocative images rated as indistinguishable from authentic photographs [86].
Pre-estimating item parameters (§ 2.1.3)	Predict item difficulty, discrimination, and response time prior to human administration, streamlining test development.	Direct LLM prediction of IRT difficulty parameters [55]; LLM ensembles as synthetic respondents for IRT-based analyses [84].
Translation & cross-cultural adaptation (§ 2.1.4)	Automated translation and cultural adaptation of psychological instruments while preserving construct validity.	Accurate, conceptually equivalent cross-cultural adaptations [128]; LLM translations outperforming traditional methods and approaching expert-level quality [66].
LLM-based response simulation (§ 2.2)	LLMs prompted to respond as specific individuals or groups across psychological measures, enabling rapid prototyping and rare scenario exploration.	Persona-prompted LLMs maintaining behavioral consistency across contexts [2]; GPT-4 predicting 91% of treatment effect variation across 70 experiments [3].
Estimating psychological states and traits (§ 2.3)	Measure psychological constructs from unstructured data (social media, speech, video) using multimodal FMs.	Personality estimation from social media posts [106]; Multimodal trait estimation integrating dialogue, speech, and video [75].
Adaptive assessment (§ 2.4.1)	Dynamic item selection and cognitive diagnosis that adapts to individual responses in real-time, improving measurement efficiency.	TestAgent analyzing open-ended responses for adaptive testing [160]; Multi-agent systems for optimal question selection [21].
Interactive assessment (§ 2.4.2)	Transform static assessments into engaging dialogues or game-like experiences while maintaining psychometric rigor.	PsyDI conducts psychological profiling through free-form conversations [78]; PsychoGAT embeds validated scales within interactive fiction games [155].
Score interpretation & intervention (§ 2.5)	Agentic systems that integrate assessment findings and generate personalized recommendations for treatment or intervention.	Psy-LLM conducts guided interviews and generates personalized suggestions [68]; PsychBench challenges models to prioritize medications and recommend treatment plans [82].

creative assessment, and the item qualities are shown to be retained [60, 73]. Large-scale evaluations in higher education demonstrate that LLM-generated items can achieve psychometric performance comparable to expert-created questions for standardized examinations [54]. Further advancement introduces a comprehensive framework leveraging network psychometrics to generate items that are simultaneously diverse, theoretically grounded, and structurally robust [119].

Beyond text-based items, FMs enable the generation of emotionally evocative visual stimuli for psychological research. Orchestrating LLMs and vision FMs, researchers can create images that are rated as nearly indistinguishable from authentic photographs, demonstrating the potential of high-fidelity synthetic visual stimuli [86].

This progress extends to the evaluation of AI itself. As LLMs exhibit increasingly human-like characteristics, researchers have begun evaluating the psychological traits of these models using items that are pre-authored or dynamically generated by LLMs [23, 117, 158]. Parallel efforts focus on training LLMs to serve as sophisticated item generators, capable of producing novel and discriminating items that adapt to or extrapolate from specified difficulty parameters [57, 141].

2.1.3 Pre-estimating item parameters

Pre-estimating the psychometric properties of test items—such as Item Response Theory (IRT) difficulty, discrimination, and response time—prior to human administration can streamline test development and reduce piloting demands. LLMs facilitate this process and enable efficient, low-cost screening of large pools of candidate items.

Two primary strategies have emerged for leveraging LLMs in this context. The first involves using LLMs to directly predict psychometric properties [55] or to extract linguistic features that inform subsequent ML models [116]. The second approach treats LLMs as synthetic respondents, generating simulated response data for IRT-based analyses [135]. However, without additional training or calibration, direct LLM-based scoring is often unstable and misaligned with human data [55, 135]. Ensembles of LLMs [84] or models specifically trained for IRT alignment [121] can better approximate item parameters.

2.1.4 Translation and cross-cultural adaptation

Adapting psychometric instruments across cultures and languages requires preserving validity and equivalence. Recent advancements include AI-assisted translation, where systems evaluate and refine human-generated translations [45], as well as fully automated translation by LLMs [124, 128]. Notably, expert-blind evaluations indicate that LLM-based translations can outperform traditional methods and approach expert-level quality [66]. Nevertheless, translation performance declines substantially for low-resource languages and is influenced by the genetic similarity between source and target languages [66]. Approaches like multicultural data augmentation and culture-aware fine-tuning offer promising solutions to these challenges [74].

2.2 LLM-based response simulation

Psychometric data collection traditionally relies on recruited participants and repeated field trials—a process that is slow, expensive, and limited in scope. Emerging research therefore explores using LLMs to simulate human respondents across diverse psychological contexts. This approach promises scalable complement to human data collection, enabling researchers to probe rare scenarios, explore counterfactual design choices, and rapidly iterate on instrument development. The premise is straightforward: LLMs can be prompted with contextual cues and instructed to respond as specific individuals would across psychological measures. However, the fundamental question remains whether synthetic responses can meaningfully substitute for human psychological data.

Current evidence presents a mixed picture. On one hand, models demonstrate remarkable capabilities in maintaining behavioral consistency across contexts. Persona-prompted LLMs can produce responses that align with established personality structures, sustain stable trait-like patterns across repeated assessments, and exhibit theoretically coherent relationships between different psychological constructs [2, 83, 122]. When scaled to population levels, synthetic respondents can approximate subgroup-consistent response patterns and recover broad conceptual landscapes observed in human samples, particularly when conditioning cues are carefully calibrated to empirical priors [4]. Recent

large-scale evaluations provide encouraging evidence [3]: GPT-4 predicts approximately 91% of the variation in average treatment effects across 70 preregistered, U.S.-representative social-science experiments [48]; fine-tuned LLMs can outperform existing cognitive models in predicting individual decision making [12]; and large-scale agent simulations (N=1,052) built from interview transcripts successfully predict participants' survey responses with 85% of the accuracy [103].

However, research reveals systematic limitations that challenge the psychometric utility of this methodology [3]. First, *diversity*: LLM outputs are often too generic or stereotyped, failing to capture the rich variation observed in human populations. For example, LLMs produce homogeneous and narrowly clustered responses in 11-20 money request game, while human participants show a more diverse response distribution [35]. Second, *demographic bias*: models exhibit systematic inaccuracies for certain demographic groups, harmfully misportraying and flattening the representations of identity groups [147]. Third, *sycophancy*: models attempt to please users or provide socially desirable responses [120, 130]. For example, LLMs tend to respond affirmatively to both original and reverse-coded items [136]. Fourth, *alienness*: results may appear human-like on the surface but arise from fundamentally non-human mechanisms, leading to empirical artifacts such as over-correlated personality dimensions [108]. Fifth, *generalization*: accuracy deteriorates in out-of-distribution settings involving novel measures, instruments, or populations.

These limitations are compounded by the brittleness of simulated responses to seemingly minor variations in prompting or model configuration. Studies document substantial shifts not only in response means but also in correlation structures and psychometric properties when prompts are reworded or when different model versions are employed [13, 125]. This sensitivity undermines the stability that is fundamental to psychological measurement and raises questions about the replicability of findings based on synthetic data. LLMs also exhibit concerning response sets. When responding to multiple-choice questions with minimalist prompts, simulations display systematic ordering biases (e.g., a preference for the first option) and labeling biases (e.g., a preference for option "A") that, after adjustment, result in uniformly random option distributions [29].

The fidelity and utility of response simulations appear to depend on several methodological and experimental factors. *Prompting strategies* play a crucial role: specifying clear, detailed personas with rich contextual information (e.g., backstories and values) substantially improves simulation accuracy compared to simple or abstract prompts [4, 51, 80]. Context-rich approaches such as incorporating hour-long interview transcripts can better anchor simulated individuals [103]. Evidence suggests that framing the task as expert forecasting, rather than role-playing, is beneficial [48]; instead of prompting, "You are a person with X characteristics," researchers can ask, "What proportion of people with X characteristics would respond Y?" This "LLM-as-expert" framing can help counter sycophancy and social desirability bias [3].

Test format also influences simulation quality. LLMs perform better on structured, judgment-based tests with clear cues (e.g., multiple-choice questions, scenario ratings) [83] and should be avoided for unstructured or projective tests that require deep, embodied understanding or creative expression [35]. *Model selection* matters as well: base models that have not undergone extensive safety alignment better capture the full distribution of human responses, as safety alignment trains models to produce socially desirable and normative responses, suppressing heterogeneity and inducing bias [3, 22, 120]. Fine-tuning models on curated social-science datasets can mitigate assistant-style biases and improve simulation accuracy [3, 80]. Finally, the *underlying task* shapes appropriateness: simulations are most suitable for early-stage prototyping (e.g., generating item candidates, testing questionnaire wording, exploring theoretical boundaries) and non-interactive, judgment-only uses [28], while strategic or interactive tasks and socially sensitive topics pose greater challenges [35]. Direct simulation of minority or marginalized groups as substitutes for real human data raises significant ethical concerns and validity issues [13].

Looking ahead, LLM-based social simulations are already useful for pilot studies, exploratory research, and sensitivity analyses to guide study design before collecting human data [3]. As accuracy improves, these simulations may enable exact replications of prior studies and, eventually, large-scale or counterfactual experiments that would be impractical or impossible with human participants. Realizing this potential requires ongoing validation, standardized evaluation protocols, and careful mapping of where simulations succeed or fail. For more comprehensive discussion, see the referenced position paper [3].

2.3 Estimating psychological states and traits

Psychometric scoring transforms responses to measurement items into numerical indicators of underlying latent constructs. Traditionally, scoring is based on predetermined keys applied to responses from standardized and validated instruments, where each item has an established relationship to the construct. With digital-behavioral data, the functional equivalent of "items" may be traces from everyday activity—such as language use, speech patterns, or video-based behavioral cues—that can then be linked to constructs through ML models. FMs extend this paradigm by leveraging powerful multimodal representations and mapping complex, unstructured inputs onto quantitative construct estimates.

Broadly, three methodological families have emerged. First, LLM-as-a-judge approaches prompt LLMs to read text (for example, social media posts, human-AI conversations, and interviews) and output state or trait estimates [106, 107, 170]. Zero-shot estimation yields modest but meaningful convergent validity with self-reports on personality tests (typical $r \approx 0.2-0.3$), which improves when interactions are designed to elicit diagnostic content (mean $r \approx 0.44$ in elicitation-focused chats) [107]. The methods can benefit from few-shot prompting (incorporating a few solution examples in prompts for LLMs to learn in situ), reasoning-enhancing prompting [104, 123], and fine-tuning on labeled datasets [152].

Second, representation-based approaches leverage FM embeddings as features for downstream ML-based classification or regression. This hybrid strategy yields lower prediction error for personality traits [90] and mental health [114] compared to zero-shot LLM inference. Notably, supplementing these embeddings with additional linguistic features does not further enhance performance, suggesting that FM representations inherently capture key linguistic characteristics (e.g., those compiled by LIWC [16]) without the need for explicit feature engineering.

A third line of research seeks to align the reasoning processes and outputs of LLMs with traditional psychometric measures, thereby strengthening both theoretical foundations and empirical validity. In one approach, LLMs are prompted to rate established items based on the content to be assessed (e.g., social media posts), and these ratings are subsequently aggregated to yield construct-level scores [127, 154, 156]. This procedure enables the calculation of internal consistency in a manner directly analogous to conventional psychometric assessments. Alternatively, LLMs can be employed to extract novel item-like statements (e.g., generating statements such as "I love parties" from posts expressing enthusiasm for social gatherings) [158]. The resulting measurements are then validated against external criteria, such as established theoretical frameworks, trait structures, and prior computational methods (e.g., theory-driven dictionaries).

The applications of FMs in state/trait estimation extend beyond textual content. Multimodal FMs integrate information from multiple cues, such as dialogue transcripts, speech, and video, to improve prediction [75, 167]. After task-specific fine-tuning, FMs are also able to capture subtle cues from facial expressions, enabling emotion recognition [165] and the diagnosis of depression [59].

2.4 Adaptive and interactive assessment

Traditional psychological assessment relies on standardized instruments that present the same questions to all individuals, regardless of their unique characteristics or response patterns. This one-size-fits-all approach can be inefficient; it potentially over-tests some individuals while under-sampling others, and it often fails to engage participants meaningfully. FMs are now enabling a fundamental shift toward dynamic, personalized assessment systems that adapt in real-time to individual responses and contextual factors.

2.4.1 Adaptive assessment

Adaptive assessment tailors the assessment experience to each individual. These systems operate through three interconnected components that work together to optimize measurement efficiency and accuracy. First, cognitive diagnosis algorithms continuously estimate the target psychological construct based on accumulated responses, updating their understanding as new information becomes available. Second, question selection algorithms identify the most informative items to present next, based on the current estimates. Third, comprehensive item banks provide the reservoir of questions from which selections are made.

FMs promise to enhance each of these components. For cognitive diagnosis, LLM agents can now analyze responses to open-ended questions, detecting nuanced patterns and anomalies that were previously inaccessible to automated systems [160]. This capability extends adaptive assessment beyond traditional multiple-choice formats to include rich, naturalistic responses. For question selection, multi-agent systems can summarize response histories, reason about optimal next questions, and dynamically refine their selection strategies [21]. The question banks themselves are expanding through automated item generation, which can scale the creation of passages, questions, and conversational scenarios to unprecedented levels (see § 2.1.2). These advances collectively enable more efficient and personalized assessments.

A parallel development involves using adaptive assessment principles to evaluate LLMs themselves. Often under IRT frameworks, researchers dynamically calibrate item difficulty based on AI performance and weighting items by their inferred difficulty, aiming to achieve accurate evaluations with smaller test sizes and more discriminative items [43, 69, 110].

2.4.2 Interactive assessment

Beyond adaptation, FMs enable entirely new forms of interactive assessment that transform the assessment experience from a passive question-answering session into an engaging dialogue. This shift addresses longstanding critiques of traditional assessment methods, which are often characterized by their static nature, lack of personalization, and limited participant engagement. Moreover, it democratizes access to sophisticated psychological evaluation, making assessment tools available beyond traditional clinical or research settings.

Current approaches to interactive assessment fall along a spectrum of structure and openness. At one end, frameworks transform established psychological scales into personalized, engaging interactive fiction games. This approach maintains the psychometric rigor of closed-ended assessment while dramatically enhancing user experience [155]. At the other end, researchers employ conversational AI systems that measure psychological constructs through free-form, multi-turn conversations. This approach allows for comprehensive profiling while maintaining the natural flow of human dialogue [78].

Two approaches balance comprehensiveness and nuance (free-form dialogue) against the higher reliability and validity (structured, game-like test formats). Encouragingly, early evidence suggests that, regardless of the specific approach, participants consistently report higher engagement and satisfaction with interactive assessment systems compared to conventional tests [155].

2.5 Score interpretation and intervention

The value of psychometric assessment extends beyond generating numerical scores to encompass the interpretation of results and recommendation of actionable interventions. Emerging attempts orchestrate FMs to build agentic systems that can conduct guided interviews, integrate findings across items and scales, and generate personalized suggestions for symptom management or referral [50, 68]. Recent benchmarking tasks further challenge models to prioritize medications or recommend treatment plans based on detailed clinical vignettes, underscoring growing expectations that AI should facilitate not only diagnosis but also care planning [82]. However, these approaches remain in their infancy, with limited validation of their clinical utility and significant ethical concerns about AI systems’ making treatment recommendations without human oversight.

3 Enhancing Foundation Models for psychometrics

FMs demonstrate strong capabilities out of the box, but their performance can be further improved through enhancement strategies. This section presents methods for optimizing FMs in psychometric contexts.

3.1 Prompt engineering

FM performance is inherently dependent on the input context [94]. Thoughtful prompt engineering can unlock their superior psychometric utility. Broadly, prompts in psychometrics serve as controllers

Table 2: Overview of enhancement strategies for Foundation Models in psychometrics.

Enhancement Strategy	Technique & Key Benefits	Representative Examples
Prompt engineering (§ 3.1)	<i>Prompts as controllers:</i> Enforce constraints on FM outputs (e.g., item attributes, construct definitions, scoring rubrics).	Specifying item attributes (e.g., topic, difficulty) for item generation [6]; Incorporating analytical rubrics to improve scoring accuracy [150].
	<i>Prompts as scaffolds:</i> Provide structured guidance or intermediate steps to improve model performance.	Guiding models with psychological questionnaires as a chain-of-thought (PsyCoT) to improve personality detection from text [156].
	<i>Prompts as levers:</i> Actively manipulate and probe FM behavior for role-playing, improving simulation fidelity, and enhancing measurement reliability and validity.	Prompting LLMs to act as a "psychologist" and "game designer" to convert scales into interactive fiction games [155]; Using detailed, behaviorally specific descriptions to improve response simulation fidelity [56].
Fine-tuning (§ 3.2)	<i>For semantic embeddings:</i> Match cosine similarity of item embeddings with empirical item-item correlations.	Fine-tuning embedding models using IPIP dataset to make the cosine similarity between item embeddings match empirical item-item correlations [151].
	<i>For item generation & parameter estimation:</i> Control construct fidelity and difficulty, screen generated items, and pre-calibrate item parameters.	Fine-tuning on the IPIP dataset for better personality-specific item generation [49]; Training FM evaluators to screen generated items [40].
	<i>For response simulation:</i> Calibrate models at individual, group, or distributional levels to predict responses.	Training models with distributional calibration to output the full distribution of population responses, conditioned on group characteristics [20].
	<i>For measuring psychological properties:</i> Predict psychometric targets from free-form content (e.g., text, multimodal data) using end-to-end or hybrid approaches.	End-to-end fine-tuning on counseling dialogues to predict psychometric scores [154]; A hybrid pipeline using FM embeddings from social media to estimate personality [90].
	<i>For adaptive assessment:</i> Train parameter estimators and question generators using calibrated IRT parameters as supervision signals.	Using reinforcement learning to train an item generator that is rewarded for producing items matching target IRT parameters [141].
Agentic AI (§ 3.3)	<i>Single-agent systems:</i> Use tools, interact with the world, and adapt behavior based on feedback.	A single agent adapts to user answers by selecting items and updating profiles iteratively [78]; Using Retrieval-Augmented Generation (RAG), where the agent retrieves expert reasoning for personality detection [79].
	<i>Multi-agent systems:</i> Coordinate multiple specialized agents to collaboratively achieve complex psychometric outcomes.	"TestAgent" uses multiple agents for adaptive assessment, including selecting questions, eliciting answers, and generating diagnostic reports [160].
	<i>Multimodal orchestration:</i> Chain and integrate FMs across modalities (text, audio, visual) to enable more versatile capabilities.	A workflow combining language and visual FMs generates emotionally evocative images, with agents responsible for prompt formation, description diversification, and image synthesis [86].

(to enforce constraints), scaffolds (to elicit capabilities), and levers (to shape synthetic respondent behavior).

Prompts as controllers explicitly instruct and constrain FM outputs for specific psychometric purposes. In automatic item generation, practitioners can effectively manage item attributes by clearly specifying the topic, difficulty level, item format, and design features within their prompts [6, 11, 95]. When combined with carefully chosen, authoritative examples ("few-shot" learning), this approach allows for more precise control over both the difficulty and stylistic characteristics of generated items [6, 99]. For measurement tasks, practitioners achieve better results by providing comprehensive, operationalized definitions of the constructs they wish to assess, rather than relying solely on construct names [117, 143]. Similarly, incorporating high-quality analytical rubrics that mirror human grading logic enhances the scoring accuracy of LLMs [150].

Prompts as scaffolds provide structured guidance or intermediate steps to support and enhance the model's reasoning, ultimately yielding better psychometric outcomes. AI practitioners typically employ scaffolding prompts to plan steps (planning), think aloud (reasoning), check the model's own output (self-reflection), and iteratively improve generation (self-improvement). For example, a popular and generic approach is to instruct models to "think step-by-step," incorporating deliberative reasoning (e.g., chain-of-thought) that mirrors human System 2 processing [148]. Some psychometric applications also employ this generic strategy [79, 170], often resulting in notable improvements. Others utilize more domain-specific scaffolds [88, 156]. For example, researchers introduce using psychological questionnaires as chain-of-thought (PsyCoT), which guides models through a structured assessment process: first generating estimated scores for individual scale items, then aggregating these scores into final construct-level estimates [156]. A similar approach explicitly instructs the LLM to reason about the emotions and emotion regulation underlying texts before outputting the detected personality traits [79].

Prompts as levers actively manipulate and probe the behavior of FMs. In interactive assessment, LLMs are prompted to take on roles such as "psychologist" and "game designer," converting scales into interactive fiction. These role-playing prompts guide both content and tone, reportedly improving engagement while preserving reliability and validity [155].

In response simulation tasks, theory-informed and role-playing prompts can improve simulation fidelity [67, 122]. For example, researchers achieve this by expanding basic personality lexicons into detailed, behaviorally specific personality descriptions, enabling LLMs to adopt diverse yet theoretically consistent behaviors [56]. More comprehensive approaches incorporate empirically derived human belief networks, aligning role-playing LLMs across 64 topics and 9 latent factors [24]. In general, providing more detailed information about the target profile (e.g., hour-long interview transcripts) can maximize simulation fidelity [4, 30, 131]. Furthermore, the framing of the simulation task itself matters: "LLM-as-expert" prompts that ask models to forecast what people would do may yield more accurate predictions than "LLM-as-subject" role-playing prompts, as discussed in § 2.2.

Prompts as levers can also improve the reliability and validity of measurements. By employing a diverse set of prompt variants and aggregating their outputs, practitioners can more thoroughly assess and improve the reliability (e.g., internal consistency) of the resulting measurements [127]. Prompt variants also help address FM biases such as option order bias by shuffling the options randomly [29]. Distribution elicitation techniques, which request models to output full response distributions or proportions rather than single responses, offer another approach to counter sycophancy and improve the validity of population-level estimates [3].

3.2 Fine-tuning

FMs are pretrained on vast datasets, enabling them to perform a wide range of tasks out-of-the-box. However, their performance on specialized tasks can often be further improved through fine-tuning, a process in which FMs are trained on smaller, task-specific datasets to better adapt to particular applications. Fine-tuning is commonly employed to enhance model performance on psychometric tasks.

When training semantic embeddings to capture psychometric relationships [32, 151], the objective is to ensure that the cosine similarity between item embeddings closely matches the empirical item-item correlations observed in human data. Typically, the embedding model takes psychometric items as input and outputs their high-dimensional representations. Its training maximizes the

correspondence between embedding similarities and ground-truth correlation matrices, often via regression-based losses. Open-Source Psychometrics Project [101] can be used to construct the training and validation dataset. Its data includes both the raw item content and the empirical correlation structure, facilitating supervised fine-tuning and robust benchmarking of the models.

In automatic item generation and parameter estimation, fine-tuning serves three main objectives. The first objective is *construct fidelity and difficulty control for item generation*. Fine-tuned models can follow detailed instructions more closely and can be adjusted to create items with specific levels of difficulty. For example, fine-tuning FMs on the International Personality Item Pool (IPIP) [39] enables better construct-specific item generation [47, 49]. Training both difficulty evaluators and question generators allows for better control over item difficulty and improves test efficiency [43, 69, 110, 142]. The second objective is *reliable and scalable screening of generated items*. Fine-tuned FM evaluators, trained to imitate human item evaluators, act as more reliable proxies to triage large candidate pools before human review [40, 99, 171]. The third objective is *accurate pre-calibration for parameter estimation*. Research fine-tunes FMs on historical responses (and their fitted IRT models) in hopes of simulating how test takers would have responded to unseen items [121, 162, 163]. As an alternative to response simulation, other approaches directly train FMs, or downstream ML predictors based on FM generated features, to predict IRT parameters from item content [19, 61, 77, 93, 116].

In response simulation, existing training approaches can be categorized by their level of calibration and optimization objective: individual-level calibration, group-level calibration by characteristics, and distributional calibration [58]. For *individual-level calibration*, researchers train models to capture and predict individual survey responses [58, 58] and behaviors in experiments [12]. These models are typically trained using standard supervised learning methods, taking as input individual behaviors or profiles and learning to predict unseen responses through imitation (e.g., maximizing the likelihood of the "true response"). For *characteristics-driven calibration*, the premise is that factors such as demographics and value preferences influence decision-making. Accordingly, models are trained to predict responses based on these characteristics [53, 111, 131]. However, this approach cannot capture intra-group variation. To address this, research on *distributional calibration* trains models to output the full distribution of population responses, with or without conditioning on group characteristics [20, 129]. These models are usually trained by minimizing the divergence between the predicted and ground-truth distributions.

Recent research advocates for individual-level modeling, as it provides more nuanced representations. In addition, while distributional data is highly demanding to obtain, it can be derived by marginalizing over group characteristics using individual-level models [131]. Taken together, these training approaches improve individual-level and group-level marginal alignment and enable some generalization across surveys, experiments, and groups. However, true generalization beyond the training data remains questionable and requires further validation [3, 25, 108, 157].

Training for measurement usually relies on datasets that pair free-form, ecologically valid content (e.g., counseling dialogues [154] and social media posts [158]) with either self-report psychometric scores or expert psychological annotations. Annotation granularity varies, ranging from item-level (e.g., associating brief textual descriptions with value valence or affective ratings) [158] to discourse-level (e.g., mapping entire blog posts to personality profiles or clinical categories) [90].

Some supervised fine-tuning approaches directly optimize FMs to predict psychometric targets from raw text, leveraging large-scale annotated corpora [154]. Alternatively, research has introduced two main hybrid pipelines. In one approach, FMs are first used to generate high-dimensional embeddings from free-form content, which are then fed into downstream ML models to estimate psychometric constructs [90]. In the other approach, theory-driven linguistic features are extracted from the free-form content, and their counts are provided as input to fine-tuned FMs for construct estimation [62]. These hybrid strategies often enhance computational efficiency and interpretability. In addition, the integration of multimodal and temporal data introduces additional technical complexity, necessitating the development of fusion architectures, specialized regression heads, and temporal modeling components [34, 75, 153].

Challenges persist despite methodological advances. Robustness to domain shift—generalization beyond the specific contexts represented in training data—remains limited. Systematic evaluation of reliability and validity at the level of established psychometric standards is rarely reported. Critically, the dependence on paired free-form content and gold-standard psychometric labels imposes

substantial costs of data collection and annotation, limiting scalability and the breadth of construct coverage.

Training for adaptive assessment typically integrates a parameter estimator and a question generator, both of which are also discussed individually above. The process generally begins by constructing psychometric targets (labels) from existing items. This involves calibrating items using response data from examinees (human, AI, or a hybrid of both) to estimate IRT parameters: primarily difficulty (b), and sometimes discrimination (a) and guessing (c), typically via expectation-maximization methods [27]. These calibrated parameters then serve as supervision signals for subsequent model training [139, 141]. As discussed in automatic parameter estimation, synthetic response patterns generated by FMs can augment or partially replace human data, thereby extending coverage and reducing costs [83, 89].

Using the constructed datasets, one can train a regression model to estimate item IRT parameters directly from item content. The model is typically optimized to minimize the mean squared error between predicted and empirical parameters, enabling accurate parameter estimation prior to field validation [141]. Building on this, a parameter-conditioned item generator can be trained to produce items that align with specified psychometric targets (e.g., difficulty). Training paradigms include both supervised learning, which encourages the generator to emulate curated item exemplars [89], and reinforcement learning, which rewards the generator for producing items whose estimated parameters (as predicted by the parameter prediction model) closely match the intended targets [141]. These two models, i.e., the parameter predictor and the item generator, can be trained sequentially or in a joint, co-evolutionary framework. The latter facilitates mutual refinement and improved alignment between psychometric calibration and item generation [57].

These training techniques are complementary and can be combined into multi-stage pipelines for advanced applications such as interactive, adaptive assessment and real-time psychometric assessment. For example, a system may integrate fine-tuned item generators, parameter predictors, and scoring models within an agentic framework to iteratively select, generate, calibrate items, and evaluate open-ended responses.

3.3 Agentic AI

Agentic AI refers to the design and deployment of multi-step, autonomous, and goal-directed AI systems capable of planning, reasoning, and collaborating to accomplish complex psychometric tasks. State-of-the-art agentic AI systems have shown promise in addressing complex scientific challenges, such as designing novel nanobodies [134], creating molecules with specific properties [161], and advancing mathematical frontiers [100]. In psychometrics, agentic AI is still in its early stages of development and is broadly explored in three ways: single-agent, multi-agent, and multimodal approaches.

Single-agent systems can use tools, interact with the external world, and iteratively adjust their behavior based on feedback. In psychometric contexts, these agents adapt to respondents' answers, selecting items, conversing with users, and updating profiles through ongoing, iterative processes [78, 145]. Another key advantage of agentic AI is its ability to retrieve knowledge from external sources. FMs encode static knowledge in their parameters and may not meet the needs of specialized tasks, especially those requiring domain expertise. In such cases, retrieving information from external sources is essential. Retrieval-augmented generation (RAG) in agentic AI injects knowledge by retrieving expert information from external databases using semantic matching. For example, one system retrieves expert-curated reasoning processes to guide LLMs in more accurate personality detection [79]. Similarly, systems for educational question generation, improvement, and evaluation ground their outputs in domain-specific course materials retrieved from relevant sources [99].

Multi-agent systems coordinate multiple AI agents, each assigned a distinct role, to collaboratively achieve psychometric outcomes that surpass the capabilities of single models. These systems break down complex measurement tasks into specialized subtasks, with each agent contributing complementary expertise. "Generative psychometrics" illustrates this approach through a systematic three-stage pipeline for value measurement [158]. First, specialized agents extract item-like observations from free-form text. Next, additional agents measure the values embedded in each observation. Finally, an aggregation step synthesizes these individual measurements into comprehensive construct scores. TestAgent, as another example, positions LLMs as adaptive assessment experts that select

personalized questions, elicit open-ended answers, generate similar or variant questions to address label or subjective uncertainty, detect anomalies, and produce diagnostic reports [160]. Multi-agent frameworks can also efficiently build and enrich domain knowledge graphs, enabling applications such as automated hypothesis generation and scientific discovery [87, 140].

Multimodal orchestration represents another line of research that leverages the capabilities of FMs across different modalities. Research on generating emotional images develops workflows that chain together language and visual FMs in coordinated stages: first, seed prompt formation agents generate descriptions of emotional images; next, prompt set generation agents create diverse descriptions targeting specific emotions; finally, text-to-image synthesis agents produce the resulting emotional stimuli [86]. Expert review is strategically interleaved throughout this pipeline to ensure both scalability and validity. Moreover, multimodal orchestration across FMs enables the fusion of text, audio, and visual information for trait measurement in richer, more contextually grounded settings [75, 164, 166].

4 Theoretical implications of Foundation Models for psychometrics

The integration of FMs into psychometrics extends beyond practical and methodological innovation; it also opens new frontiers for psychological theories.

4.1 Reshaping psychological taxonomies

The ability of FMs to map psychological constructs into a shared semantic space, as shown in § 2.1.1, goes beyond methodological convenience; it offers a data-driven approach to the long-standing challenge of taxonomic incommensurability in psychology [70, 149]. This approach can help uncover common taxonomic issues in the field, such as the jingle fallacy, where measures with similar labels actually assess different constructs, and the jangle fallacy, where different labels are used for measures that assess the same construct. By providing a scalable way to quantify relationships between measures, FMs enable researchers to systematically identify and address these problems.

The potential theoretical implication is a paradigm shift from fragmented, redundant, and top-down taxonomies toward a more unified, parsimonious, and empirically grounded psychological taxonomy. Recent research supports the use of trained embeddings to clarify conceptual confusion in psychology, suggesting that a smaller set of constructs may be sufficient to capture meaningful distinctions in Psychology [151].

4.2 Discovering and re-evaluating latent structures

FMs can serve as powerful engines for discovering and re-evaluating the latent psychological structures. Embedding-based analyses have not only rediscovered established structures such as the Big Five personality traits, but have also provided evidence for alternative models, including two- or three-factor models for extensive personality lexicons [26]. Factor structures derived from everyday human language use have been shown to predict non-questionnaire-based outcomes better than traditional structures derived from questionnaire data [65].

These novel latent structures are uncovered by leveraging the unique advantages of FMs in ways traditional surveys cannot. FMs can analyze data in dozens of languages, process vast numbers of items and lexical descriptions, examine historical texts, operate at extremely large scales with minimal cost, and access unprompted expressions of human psychology. Empirical findings also suggest that the structures identified through questionnaire data are neither the only valid representations nor necessarily the optimal. Furthermore, the ability of FMs to model constructs across multiple modalities opens new theoretical avenues for examining the coherence (or lack thereof) between an individual's self-perception, written expression, and behavior. Taken together, FMs can serve as a computational "microscope," revealing more nuanced semantics than previous computational models, or as a "telescope," enabling the analysis of data at scale and across cultures, languages, modalities, and time.

4.3 Modeling cognition and representing populations

FMs present an unprecedented opportunity to simulate individuals and populations (§ 2.2). This potential stems from two key strengths of FMs: their versatility in generative modeling, and their remarkable ability to learn from context and absorb complex patterns within vast training data.

Traditionally, the development of psychological and cognitive theories has followed a hypothesis-driven approach: researchers formulate theoretical predictions and test them in small, domain-specific experiments. The resulting data are then used to fit simple statistical models, which are refined iteratively. The process is often limited by the scale of available data and the narrow scope of experimental designs.

Recent FMs are opening new pathways for theory development [12, 109, 131, 169]. Cognitive modeling FMs (e.g., Centaur [12]) are trained on large-scale human experimental data and capture behavior across diverse domains. Such black-box models can serve as scaffolds for extracting interpretable theories. Recent work shows that reasoning-enhanced FMs (e.g., DeepSeek-R1 [44]) can be prompted to distill interpretable theories from human experimental data [12]. While these distilled theories may not match the predictive accuracy of Centaur, they already explain human behavior better than expert-proposed theories. Using black-box Centaur as references enables theory refinement without additional domain-specific data collection. This iterative process can produce interpretable theories that fit as well as Centaur [12]. Such FM-guided theory discovery challenges the common belief that black-box models must necessarily trade interpretability for accuracy [33].

Beyond modeling individual cognition, using FMs to represent populations has significant theoretical implications. Empirical findings from response simulation studies suggest that the FM training data are inherently biased samples of human discourse. Consequently, without specialized tuning, FMs most likely do not directly mirror individual minds but instead reflect collective narratives, stereotypes, and folk theories embedded in cultures and societies [18, 126]. This perspective underscores their unique theoretical value. Vanilla FMs can be powerful tools that enable researchers to model how societal biases are encoded in language [22], trace the evolution of values and moral frameworks across historical periods (by training on texts from different eras) [112], and examine how collective beliefs and attitudes are constructed and propagated [102]. For instance, large-scale analyses of nine language models and 1.4 million online images reveal systematic age-related gender stereotypes, with women consistently represented as younger than men across occupations—a distortion absent in actual workforce demographics [42]. Such findings demonstrate the theoretical value of FMs for exposing and quantifying how societal biases are encoded in digital discourse at unprecedented scale.

4.4 LLM psychometrics

LLM psychometrics is referred to as an interdisciplinary field dedicated to evaluating, understanding, and enhancing LLMs through the application and integration of psychometric instruments, theories, and principles [157]. Studying LLMs as test subjects prompts a revisit of basic psychometric assumptions concerning subjects, constructs, and measurement methods.

Redefining the "subject" of measurement. Psychometrics was originally developed to assess humans. However, its recent and rapidly growing applications to FMs, especially LLMs, fundamentally challenges this core premise. Whereas human psychology assumes that traits such as personality are relatively stable over time, LLMs display contextually adaptive traits; they do not possess a single, enduring personality, but instead can adopt a wide range of personas depending on the prompts they receive [122]. There is a lack of consensus in how LLMs are treated in this line of research: a single LLM can be treated as an individual subject in its default state [158], as a population of subjects through diverse role-playing prompts [122], or as a monolithic entity generating stochastic, diverse responses to the same item [46]. This flexibility disrupts the traditional psychometric notion of a single test-taker with a unique set of traits. Additionally, the very act of applying psychometric tests to LLMs raises the issue of anthropomorphization. Attributing human-like consciousness or intentions to these models can mislead interpretation. As a result, psychometricians are compelled to a new theoretical frontier. Interdisciplinary efforts are needed to carefully distinguish between measuring human psychology and quantifying behavioral patterns in a non-human entity, and to design new theoretical systems for the latter as needed [157].

Re-examining the universality of psychological constructs. Research indicates that psychological constructs developed for humans may not be directly applicable to LLMs, compelling the field to consider how intelligence can be modeled beyond the human domain. Studies consistently show that human-centric theoretical models, such as the Big Five and Schwartz’s Value Theory, do not fit LLMs well [105, 136]. For instance, the clear separation of Big Five personality traits observed in humans is not reflected in LLM responses, and human value structures do not map cleanly onto LLM outputs. This mismatch necessitates a shift from relying solely on human constructs to developing constructs that are native to LLMs. Recent pioneering research addresses this gap by deriving new value and ability systems directly from the behavioral patterns of LLMs themselves [17, 159]. The newly established frameworks often indicate better validity, for example, in predicting and ensuring LLM safety [159]. Psychometrics may need to evolve from a science focused exclusively on human psychology to a broader discipline capable of measuring latent traits in any intelligent system. In doing so, theoretical models of these constructs may need to be reestablished.

Toward more ecologically valid measurement approaches. The unique properties of LLMs compel psychometrics to move beyond traditional, human-centric methods, particularly self-report inventories, toward more ecologically valid, behavior-based approaches. Unlike humans, LLMs lack subjective experience and have limited self-knowledge, so their responses to self-report items may be shaped more by statistical regularities than by introspection [130]. This leads to systematic artifacts such as option order effects [29], a preference for socially desirable answers [130], and sycophantic agreement regardless of the items [91, 136], all of which undermine the interpretability and validity of self-report data. Compounding these issues, LLMs are highly sensitive to subtle variations in prompt wording, syntax, and response format, resulting in inconsistent outputs that challenge the psychometric principle of parallel forms reliability [118]. These unique properties (or limitations in current AI training) necessitate alternative strategies that better capture the behavioral repertoire of LLMs. Approaches such as analyzing open-ended response [117], observing agentic behavior in simulated environments [168], and leveraging naturalistic interactions [158] offer more ecologically and empirically valid results. In addition, applying these approaches to LLMs is much simpler and more cost-efficient than to humans, as AI is readily accessible anytime, anywhere, to any query. Taken together, we argue that while self-reports are the gold standard in human psychometrics, ecologically valid approaches should be the go-to choice in LLM psychometrics.

5 Recommendations

The integration of FMs into psychometrics is rapidly advancing, yet consensus on best practices remains limited. To guide this integration, we offer the following recommendations for researchers and practitioners.

5.1 Conceptual and methodological rigor

FMs are tools for prototyping and hypothesis generation, not yet ready replacements for human participants. FMs do not share the cognitive mechanisms of human psychology; they are complex statistical models trained on vast datasets. Currently, treating them as direct substitutes for human participants can lead to flawed conclusions [81]. Instead, their strength lies in augmenting the research process. Use FMs to rapidly prototype instruments, generate novel hypotheses, and explore theoretical boundaries in-silico before costly human data collection [41, 140]. A staged roadmap for LLM social simulations suggests using them for pilot and exploratory studies in the near term; their use could expand to exact replications and large-scale or counterfactual studies as methods improve [3].

Methodological advances should be accompanied by rigorous psychometric validation against human data. This process requires comprehensive psychometric validation that goes beyond simply comparing mean scores. For example, researchers may assess the factor structure, reliability, validity, and measurement invariance of AI-generated data. Early, head-to-head comparisons with human data are essential before scaling up generation and experiments. Iterative validation frameworks can help continuously track methodological progress and identify limitations systematically.

Consider enhancing FM performance for psychometric applications when out-of-the-box FMs are not sufficient. As detailed in § 3, use prompt engineering to control generation, scaffold complex

reasoning, and probe model behavior. When data is available, fine-tuning can adapt FMs to specific domains. For complex applications, consider agentic AI architectures that orchestrate multiple models or tools.

5.2 Robustness and reliability

Design for and test against perturbations and response sets. Findings based on FMs can be brittle. Research shows that FMs (1) can be sensitive to variations in prompts, languages, and task formulations [117, 118]; (2) exhibit response set biases, such as preferring option A in multiple-choice questions [29], favoring confirmatory responses [91], and giving generally high or low scores regardless of input [158]; and (3) are partially driven by social desirability [92, 130]. To ensure robust and reliable results, systematically compare outputs across multiple models and rigorously test sensitivity to artifacts. Use perturbation tests (e.g., paraphrasing prompts or altering response option order) to evaluate result stability. Assess self-consistency by generating and comparing multiple outputs from the same model. One should maintain vigilance for artifacts throughout study design and interpretation.

Address potential data contamination and the temporal context of FMs. FMs may have been exposed to evaluation tasks and data during training, which can artificially boost performance and undermine validity. Therefore, it is important to guard against data contamination to ensure robust findings. For example, when using FMs for experimental replication, it is ideal to use results from studies published after the model’s knowledge cutoff. Conversely, when leveraging LLMs’ encoded knowledge, one should recognize the temporal limitations imposed by their training data.

Ensure transparent and comprehensive reporting. Reproducibility is a cornerstone of scientific progress. When reporting results from FM-based research, provide complete documentation of your methods. This should include the models used, model versions, full prompts, hyperparameters such as decoding settings, and other implementation details. To support comprehensive reporting, [Box 1](#) offers a checklist that supplements traditional guidelines for psychological research by addressing considerations unique to FM-based methods.

5.3 Responsible and ethical integration

Make fairness a core principle. Ethical use of FMs in psychometrics requires a strong commitment to fairness. Researchers should conduct statistical analyses for differential item or algorithmic functioning [133]. Audits for bias should be implemented by evaluating FM-generated items and responses across diverse demographic groups to identify and mitigate biases [37, 137]. When necessary, apply bias mitigation strategies [137].

Foster interdisciplinary collaboration. Progress in this rapidly evolving field depends on close collaboration between AI experts and psychologists. AI researchers contribute technical knowledge, while psychologists ensure applications are grounded in sound measurement and ethics. Such partnerships are vital for developing tools that are reliable, valid, and fair.

Adopt a socio-technical perspective on governance. Treat FM-based psychometric tools as socio-technical systems when deploying them at scale (e.g., personality tests on websites). Carefully assess potential risks and implement holistic governance that accounts for societal impacts, following established standards [10].

Box 1 | Additional reporting checklist for FM-based psychometric research

Model information

- ☐ Model name(s), version(s), and release date(s)
- ☐ Model provider/source
- ☐ Model access method (API, local inference)
- ☐ Knowledge cutoff date
- ☐ Justification for model selection criteria

Experimental design

- ☐ Complete prompts and instructions (including system messages)
- ☐ Hyperparameters (temperature, top-p, max tokens, etc.)
- ☐ Number of runs/samples generated per condition

Validation and quality control

- ☐ Psychometric validation methods and results
- ☐ Mitigation strategies for data contamination
- ☐ Perturbation tests and sensitivity analyses
- ☐ Response quality filtering and exclusion criteria

Ethical and fairness considerations

- ☐ Demographic groups and characteristics examined
- ☐ Bias detection methods and statistical tests employed
- ☐ Bias mitigation strategies implemented (if applicable)
- ☐ Potential risks and harms associated with FM use
- ☐ Intended use and potential misuse considerations

Reproducibility

- ☐ Code availability
- ☐ Data availability (synthetic data, prompts, model outputs, etc.)
- ☐ Computational environment and dependencies
- ☐ Reproducibility measures
- ☐ Estimated computational costs and resources required

Limitations and scope

- ☐ Key limitations of the FM-based approach
- ☐ Scope and applicability of the findings

6 Summary and future directions

The integration of FMs into psychometrics promises a paradigm shift. It broadens the scope of what can be measured, introduces new methodologies for measurement, enables assessments across diverse settings, and reduces associated costs. Psychometrics-specific enhancement strategies further improve both the theoretical and empirical validity of this integration. On the theoretical front, FMs facilitate data-driven taxonomic clarification, the discovery of novel latent structures, streamlined theory development, and the emergence of a dedicated field of LLM psychometrics.

Research at the intersection of FMs and psychometrics is expanding rapidly, yet the field remains in its infancy. Several key scientific inquiries remain underdeveloped. First, future research should develop broader, enhanced, and more integrated applications of FMs in psychometrics. This includes designing agentic AI for adaptive testing and instrument creation, building multimodal interactive assessment systems for more engaging and valid assessments, and calibrating LLMs for individualized in-silico experiments. Second, the adaptation and enhancement of FMs for psychometric applications demands ongoing interdisciplinary collaboration. Current approaches are still limited in their generality and dependency on heavily annotated data. State-of-the-art AI methods (e.g., self-evolving algorithms) are yet to be leveraged for this purpose. Third, the theoretical impact of FMs in psychology requires further study. FMs could enable unified, data-driven taxonomies of psychological constructs and accelerate theory discovery by extracting interpretable models from large datasets. Fourth, LLM psychometrics is an emerging field that seeks rigorous measurement of analogous psychological traits

in LLMs. Interdisciplinary collaboration is needed to establish theoretical foundations, define AI-native constructs, develop valid behavior-based assessment methods, and explore algorithms to steer AI character. Finally, ethical and robust deployment of these tools is essential. Future work should develop scalable methods to ensure FM consistency and alignment, establish reliable and robust findings, and audit and mitigate algorithmic bias. Building a trustworthy science of FM-augmented psychometrics requires establishing standards, benchmarks, and platforms for transparency and reproducibility.

Collectively, the integration of FMs promises a transformative chapter for psychometrics. Realizing this potential will require interdisciplinary efforts to advance the methodological, theoretical, and ethical frontiers of the field.

A Terminologies

Box 2 | Key AI terminology

Foundation Models (FMs) — Large-scale, deep-learning models trained on vast amounts of data, which can be used for and adapted to a wide range of downstream tasks.

Large Language Models (LLMs) — A type of FM specifically trained on massive text datasets to understand, generate, and process human language. Examples include GPT-4 [1] and DeepSeek [44].

Vision Foundation Models (VFMs) — A type of FM specifically trained on massive datasets of images and videos to understand, generate, and process visual information.

Generative AI — A class of AI models, including many FMs, that can create new content, such as text, images, or code, rather than simply analyzing or classifying existing content.

Semantic Embeddings — Numerical vector representations of words, sentences, or other content in a high-dimensional space. The geometric distance between vectors in this space reflects their semantic similarity, allowing models to capture nuanced meaning.

Zero-shot, Few-shot, and In-context Learning — The ability of FMs to perform tasks without any task-specific training (zero-shot) or with only a few examples provided in the prompt (few-shot). This learning happens "in-context" during a single interaction.

Prompt Engineering — The practice of carefully designing input prompts to guide FMs toward producing desired outputs. It is a key technique for controlling and optimizing the behavior of these models without retraining them.

Retrieval-Augmented Generation (RAG) — A technique that enhances FMs by allowing them to retrieve information from an external knowledge base (e.g., a collection of documents) before generating a response. This helps to ground outputs in factual, up-to-date information.

Supervised Learning — A fundamental machine learning paradigm where a model is trained on a dataset containing input-output pairs (labeled data). The model learns to map inputs to outputs, enabling it to make predictions on new, unseen data.

Reinforcement Learning (RL) — A machine learning paradigm in which an agent interacts with an environment by taking actions and receiving feedback in the form of rewards or penalties. Through repeated trial and error, the agent learns a policy that maximizes cumulative reward over time.

Alignment — The process of ensuring that an AI system's goals and behaviors are consistent with human values and intentions. Techniques like Reinforcement Learning from Human Feedback (RLHF) are used to align models, making them more helpful, harmless, and honest. In particular, safety alignment ensures the model avoids harmful or biased outputs and adheres to ethical guidelines.

Fine-tuning — The process of further training a pretrained FM on a smaller, task-specific dataset. This adapts the model to a particular application, often improving its performance and alignment with desired behaviors.

Chain-of-Thought (CoT) Prompting [148] — A prompt engineering technique that encourages an LLM to explain its reasoning process step-by-step before giving a final answer. This often improves performance on complex reasoning tasks by mimicking a more deliberate thought process.

LLM-as-a-Judge — A method where an LLM is prompted to act as an evaluator or "judge" to score, rate, or classify content (e.g., text, responses) based on a given set of criteria or rubrics. This is often used for automated assessment and data annotation.

Agentic AI — AI systems designed to be autonomous and goal-directed. They can plan, reason, use tools, and collaborate with other agents to accomplish complex tasks, often involving interaction with an external environment.

Multi-Agent Systems — A subfield of AI that involves the design and coordination of multiple autonomous agents to solve problems that are beyond the capabilities of a single agent.

Box 3 | Key psychometric terminology

Psychometrics — The scientific discipline concerned with the theory and technique of psychological measurement. It provides rigorous frameworks for quantifying latent psychological constructs like abilities, personality, and attitudes.

Latent Psychological Constructs — Unobservable psychological attributes, traits, or abilities that are inferred from observable behaviors, such as responses to questionnaire items.

Reliability and Validity — Two fundamental criteria for evaluating the quality of a measurement instrument. Reliability refers to the consistency and stability of measurement results over time, across items, or between raters. Validity refers to the extent to which the instrument accurately measures the construct it is intended to measure.

Instrument — A structured tool used to obtain observable indicators of a construct. In psychology, instruments include questionnaires, tests, or interview protocols comprising items designed to elicit responses that reflect underlying traits or states.

Item — A single, discrete prompt or question within an instrument that elicits an observable response indicative of the target construct. Items may be statements, questions, tasks, or stimuli with specified response formats (e.g., multiple choice, Likert agreement, open-ended).

Scale — The ordered set of response options (and the associated scoring rules) by which item responses are quantified and combined. A scale defines how raw responses map onto numerical values (e.g., Likert-type 1–5 agreement), how items are summed or averaged, and the level of measurement (nominal, ordinal, interval, ratio) used to represent the construct.

Item Response Theory (IRT) — A family of statistical models that link individual item responses to the underlying latent trait being measured. IRT models can estimate item properties such as difficulty (how much of the latent trait is needed to endorse the item) and discrimination (how well the item differentiates between individuals with different levels of the trait).

Factor Structure — The underlying dimensional structure of a set of variables or test items. Factor analysis is used to identify common factors that explain the correlations among items, providing evidence for the internal structure of a construct.

Measurement Invariance — A statistical property indicating that a measurement instrument functions in the same way across different groups (e.g., demographic or cultural groups). Establishing measurement invariance is crucial for making meaningful and fair comparisons of scores across these groups.

Jingle-Jangle Fallacies — Two common errors in psychological research. The jingle fallacy is the mistaken belief that two different things are the same because they bear

the same name. The jangle fallacy is the mistaken belief that two identical things are different because they are labeled differently.

Adaptive Testing — A method of assessment in which the selection of items is dynamically adapted to the test-taker's performance level. Items are chosen to be maximally informative based on the individual's previous responses, making the test more efficient and precise.

Differential Item Functioning (DIF) — Occurs when an item on a test has different statistical properties for different subgroups of test-takers, after controlling for the underlying ability being measured. DIF is a source of test bias that can lead to unfairness.

Internal Consistency — A measure of reliability that assesses the extent to which items on a test that propose to measure the same construct produce similar scores. Cronbach's alpha is a common estimator.

Construct Validity — The degree to which a test measures the theoretical construct it purports to measure. It is an overarching concept of validity that encompasses evidence from multiple sources, including internal structure, response processes, and relations to other variables.

Response Sets — A tendency for individuals to respond to test items in a systematic way that is unrelated to the item content. Examples include acquiescence bias (yea-saying).

Social Desirability Bias — A systematic tendency to provide responses that present oneself favorably according to perceived social norms, regardless of true feelings or behaviors.

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